MIND OVER MATTER

Issue 3, November 2025

Cognitive Load Theory

Not all demands on working memory are created equal.
Cognitive Load Theory (CLT): The total mental effort for learning is divided into three key parts:

Extraneous Load:

Due to poorly designed instruction, there is unneeded complexity that does not help learner reach the goal. It takes up valuable working memory space without contributing to learning.

Examples:
Confusing slide layout
Redundant info (text read aloud verbatim)
Integrating separate sources (text and image on different slides)

Goals: Reduce it. Eliminate instructional clutter to allow our students to focus on what matters most.

Intrinsic Load:

The inherent difficulty of the material. It is based on element interactivity - the number of elements that must be processed simultaneously. High interactivity = High intrinsic load.

Examples:
Learning a complex procedure (long division, steps of Krebs Cycle, long term causes of WWII)

Goal: Manage it, Use sequencing and chunking to break down complex material into smaller, manageable units. This process can tap into our pillars of spacing and interleaving!

Germane Load:

This is the "good" cognitive load. It is the effort directly related to schema construction, the process of connecting new info to existing knowledge. This is what leads to long-term learning

Examples:
Comparing & contrasting two concepts
Explaining an idea in own words
Applying knowledge to problems

Goal: Optimize it. Free up working memory from extraneous load, ensuring that the learner has resources left for this critical step.

Actionable Integration into the Classroom: By understanding how the brain processes information, we can design lessons that are both challenging and efficient. The table below outlines four key principles for optimizing your instructional design:

Coherence Principle (Reducing Extraneous Load) →	Prune and simplify all materials. Remove decorative visuals, unnecessary text, and redundant audio/video. →	Prevents working memory from being spent on irrelevant details. The "workbench" only has the necessary tools.	01
Modality Principle (Reducing Extraneous Load) →	Present related visuals and words simultaneously using both channels: an image on the screen and related narration/explanation (not text on the screen). →	Distributes the load across two different working memory processors (visual and auditory), effectively doubling capacity and reducing redundancy.	02
Segmenting Principle (Managing Intrinsic Load) →	Break down complex lessons into small, discrete, and self-paced chunks. Pause after each segment for a quick knowledge-check or activity. →	Reduces the number of interactive elements processed at any one time, preventing overload and allowing for better schema formation.	03
Worked Example Effect (Optimizing Germane Load) →	Instead of always starting with a complex problem, present a fully solved example first, followed by a partially completed problem, and then a novel problem. →	Provides a schema to follow, drastically lowering extraneous load for novice learners. This frees up resources for them to process why the steps work (germane load).	04